

Fairness interventions and case shift

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Screening applications, including welfare fraud detection, credit scoring, and criminal risk assessment, are among the highest-stakes settings for algorithmic fairness and share a structural property. Since outcome labels are available only for cases selected by a prior screening process, the training distribution fundamentally differs from the deployment population. Fairness interventions are calibrated on such training data but must hold for the population. Here, we analyze the implications of this mismatch for the effectiveness of fairness interventions and identify two failure modes. The first is the non-identifiability of fairness measures. Most standard fairness measures require confusion matrix entries that are structurally absent under selective labeling and cannot be estimated regardless of sample size. The second is disparity reversal. Selective labeling distorts class prevalence in the training corpus relative to the population. When the model is additionally differentially accurate across demographic groups, a fairness intervention that achieves parity at the training prevalence can reverse the disparity at deployment. We derive conditions under which reversal occurs and show that non-identifiability prevents its detection on available data. We validate both failure modes against a documented case of welfare fraud detection by the City of Amsterdam, in which an extensive fairness intervention corrected for disparity on the evaluation corpus but reversed it in a live pilot. We recommend that only identified measures should guide fairness interventions, that differential accuracy should be minimized or explicitly accounted for, and that the prevalence mismatch between the corpus and the deployment population should be corrected before optimizing models for fairness.

Keywords: algorithmic fairness, selective labeling, case shift, fairness measures, disparity reversal, welfare fraud detection

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1 Introduction

Algorithmic fairness is now a mature field [6, 23]. Much of recent work has rightly emphasized that the hardest remaining challenges are not technical but social, legal, and institutional [41, 72]. We do not dispute this, yet we show that a class of technical issues in the most consequential fairness applications remains largely unaddressed.

Many algorithmic fairness applications, including welfare fraud detection [15, 37, 42, 76], credit scoring [8], criminal risk assessment [5, 34], or automated shortlisting of job applicants [30], share a structural property. Ground-truth outcomes are known only for cases that the process selected for investigation or approval. A loan applicant is labeled only if approved; a welfare claimant only if flagged for investigation; a defendant for recidivism only if released on bail. The labeled corpus used for training predictive models is therefore a structurally biased sample of the incoming population. However, models trained on such corpora are aimed at working on the full

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population. We call this mismatch a case shift and show that it gives rise to two failure modes that together can explain why fairness interventions in screening applications frequently fail.

The first failure mode is non-identifiability of fairness measures. Many standard fairness measures require confusion matrix entries that are structurally absent under selective labeling, because cases on one side of the screening decision are never investigated and thus never labeled. These measures cannot be faithfully estimated from the corpus regardless of sample size, and optimizing a fairness intervention against an unidentified measure provides no guarantee of what will happen at deployment.

The second failure mode is disparity reversal. Selective labeling distorts class prevalence in the corpus relative to the true population. Since the screening process preferentially selects cases likely to be positive, the labeled corpus is enriched for the positive class far beyond the population rate. A fairness intervention calibrated on such a corpus will not transfer cleanly to deployment. If the model is additionally differentially accurate across demographic groups, the disparity can reverse entirely.

I argue that this is what happened in the Amsterdam fraud detection system [3, 15, 42]. Between 2021 and 2023, the municipality developed and briefly deployed a system to prioritize welfare applications for fraud investigation, before abandoning it on grounds of discrimination. After an internal evaluation found that the original model disproportionately flagged non-Western applicants, the city applied a reweighing intervention, declared the bias corrected, and proceeded to a live pilot. The pilot showed the disparity had reversed: the system was now substantially unfavorable to local applicants. We reconstruct this reversal from publicly released confusion matrices [3, 15] and show that it was a predictable consequence of the structure of the learning task.

Our study makes four contributions: (1) we formalize case shift in screening applications and distinguish it from dataset shift and concept drift; (2) we catalog which standard fairness measures are identified under selective labeling and propose a correction that recovers identifiability for the remainder under a small randomized audit; (3) we derive theoretical conditions under which case shift produces disparity reversal even for identified measures, and show that non-identifiability prevents its detection on available data; and (4) we provide the first investigation of the Amsterdam welfare fraud detection case from the algorithmic perspective and show that the observed reversal was a structural consequence of the learning task setup.

2 Setting: selective labeling in screening applications

Screening applications are those that flag cases for further action. Actions then lead to outcomes that provide ground-truth labels. A selection mechanism S , whether a prior manual process or a previous version of the model, decides which cases to investigate. Outcomes are known only when $S = 1$; those cases receive ground-truth labels. Predictive models are typically trained on a labeled corpus and then deployed on the full incoming population. Since labeling is selective, the distribution of the training data differs from that of the deployment data.

This structure gives rise to two mandatory properties that are present in every screening application due to task construction. The first is systematically missing ground truth. Any case with $S = 0$ receives no outcome label. True negatives and false negatives are therefore structurally absent from the training corpus. An additional small audit can recover some labels from the uninvestigated pool, but such an audit is typically very small. If most of the unflagged cases were investigated, the screening mechanism would lose its purpose.

Given this structure imposed by the task setting, Figure 1 illustrates the pattern of observable and missing confusion matrix entries. Panel (b) indicates the structurally missing part as compared to uniform sampling.

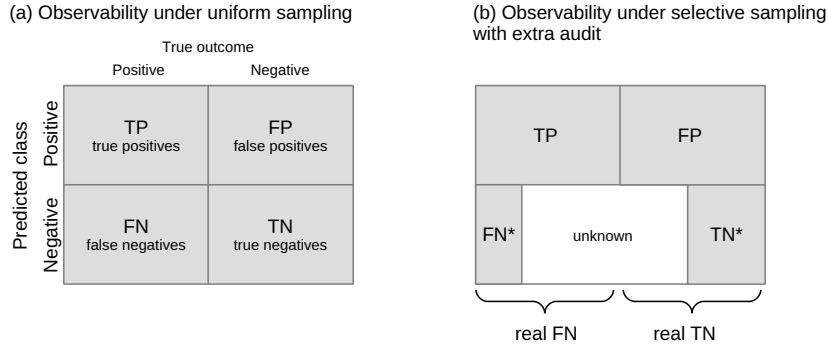


Fig. 1. Confusion matrix entries observable (shaded) and missing (white) under (a) uniform sampling and (b) selective labeling with a small additional audit *.

The second structural property is case shift. Formally, let $y \in \{0, 1\}$ denote the true outcome (1 = fraud), $a \in \{0, 1\}$ a the group membership indicator (protected attribute), and X the feature vector. Let $\mathcal{D}$ denote the true distribution over (X, a, y) among all incoming applicants, and let the training corpus follow $\mathcal{D}(\cdot \mid S = 1)$.

Definition 2.1 (Case Shift). Case shift occurs when $\mathcal{D}_{\text{train}} \neq \mathcal{D}$ and the discrepancy originates from the selection mechanism S being a non-uniform selection from the population.

Case shift is always present in screening applications because S is not independent of y . A selection mechanism with no predictive value for y would serve no purpose. Thus, case shift is endogenous to the task and is distinct from dataset shift [68] or concept drift [40], where the population $\mathcal{D}$ changes externally to the task. Here, the population does not change, while the operational process induces a change in the distribution, which is analytically predictable.

A mandatory property following from case shift is class enrichment in the training corpus. Since S has predictive value for y , positively flagged cases that receive ground-truth labels exceed the population prevalence, $P(y = 1 \mid S = 1) \gg P(y = 1)$. In the Amsterdam case that we study, for example, the estimated fraud prevalence in the population is $\approx 5\%$, while the labeled corpus has $\approx 55\%$ fraud prevalence [15].

The stronger the predictive power of the selection mechanism, the greater the class enrichment and thus the mismatch between the training corpus and the population. This distinguishes case shift from ordinary class imbalance [45], where imbalance is a fixed property of data. Under case shift, the gap between corpus and population prevalence widens as the model improves; better models thus make fairness interventions calibrated on the corpus progressively less reliable at deployment.

Beyond these mandatory properties of the learning task setting in screening applications, two further conditions are common. The first is demographic skew, which arises when the proportions of demographic groups in the corpus differ from those in the true population. In the Amsterdam case, 85% of cases in the labeled corpus are Dutch applicants, compared with an estimated 50–60% in the true applicant population [15]. This can be a consequence of the prior manual flagging process.

The second condition is differential accuracy, which means the predictions of the model are more accurate for one demographic group than another. This can arise from biased outcome labels, from proxy features that correlate

with the outcome for some groups but not others, or from demographic imbalance in the corpus, which can be further amplified by demographic skew. By standard learning-theoretic arguments [73, 77], the more training data there are, the better the model generalizes. Where one group is underrepresented in the corpus, as in the Amsterdam case, differential accuracy across demographic groups is to be expected.

These structural properties of the task setting in screening applications give rise to two serious failure modes of fairness interventions, which we analyze in turn.

3 Failure mode I: non-identifiability of fairness measures

The algorithmic fairness literature offers many measures, each capturing a different intuition about what equal treatment means [6, 18, 61, 63, 79, 87]. Theoretical results show that many of the popular measures cannot be jointly satisfied [21, 25, 54], while some combinations are achievable [33, 85]. Our concern here is a question prior to measure compatibility. We ask which measures can actually be estimated in screening applications at all.

Due to the structurally missing entries of the confusion matrix, as illustrated in Figure 1, many fairness measures cannot be meaningfully estimated from the available data. Table 1 catalogs widely used fairness measures and indicates whether each is identified under selective labeling in screening applications such as the Amsterdam welfare case, where true and false negatives are structurally absent. Typically, each measure is computed separately for each demographic group, and a disparity is declared when the measures differ substantially across groups.

We see from the table that only Predictive Parity (Precision) and its complement, the False Discovery Rate (FDR), are identified. Statistical Parity and FP Group (the measure that Amsterdam used) are identified if the population group sizes are known. This is likely possible in practice, since the demographic composition of all incoming applications is typically known, even if the ground-truth labels are unknown for some applicants.

We can further see from the table that many popular fairness measures are unidentified. This is a striking result, casting doubt on many fairness claims in practical applications. Critically, even augmenting the corpus with a small independent audit does not directly recover identifiability, because of the unknown gap shown in Figure 1 (b). Measures conditioned on explanatory attributes [26, 88] face the same structural problems.

To illustrate why unidentified measures are problematic for fairness interventions, consider the following example. Table 2 presents a numerical illustration assuming that a selectively labeled corpus is augmented by a small audit. Suppose the population consists of 2000 people, 1000 per group, with a 10% prevalence (the proportion of fraud in the population) and no true disparity between groups. Suppose that the investigation accuracy is better than random, but different between the two groups. The investigation correctly flags fraud with a probability of 30% for group A and 10% for group B. Suppose that after the model flags cases for investigation, additional 10% cases are selected at random for checking in identical proportions for both groups, which gives samples FN and TN. The populations FN and TN are invisible to the investigator, but here we know them because we have ground truth from simulation.

Suppose the predictive model is calibrated to equalize the False Positive Rate (FPR*) on the training data. Table 2 shows the results of this fairness intervention. We see that, while the model training equalized FPR*, the true underlying FPR remained unequal. In this case, the difference in accuracy between the groups is due to differing accuracy. Examples can also be constructed in which disparity reverses, as we see later in this paper. This implies that, as the measure is unidentified, calibrating fairness intervention to it does not make sense, because the effects on the true population can be very different.

The non-identifiability of measures has direct implications for fairness interventions in practice. Several measures that have been argued to be the most normatively appropriate, including Equalized Odds [43], Equal Opportunity [43], and FPR parity [5, 21], are unidentified in screening applications. Yet these applications include welfare, access to financing, criminal sentencing, childcare, and more, and are among the most consequential algorithmic fairness applications.

We propose that the non-identifiability of fairness measures can be partially remedied by a statistical correction based on a randomized audit, in which a small fraction β of uninvestigated cases is selected uniformly at random and labeled. From the audit sample FN_{audit} and TN_{audit} can be computed and then extrapolated to the true population as follows.

Let $N = n_{\text{investigated}} + n_{\text{uninvestigated}}$ be the number of cases per group flagged by the model and left uninvestigated correspondingly. $n_{\text{investigated}} = \text{TP} + \text{FP}$. Let n_{audit} indicate the number of cases per group randomly additionally investigated for audit purposes. Let $\beta = n_{\text{audit}}/n_{\text{uninvestigated}} = n_{\text{audit}}/(N - \text{TP} - \text{FP})$. Since the audit is a uniform random sample of the uninvestigated pool, the observed counts FN_{audit} and TN_{audit} are unbiased estimates of the corresponding fractions in that pool. Scaling by the inverse sampling fraction $1/\beta$ gives unbiased estimates of the population total

$$\widehat{\text{FN}} = \text{FN}_{\text{audit}}/\beta, \quad \widehat{\text{TN}} = \text{TN}_{\text{audit}}/\beta. \quad (1)$$

Table 1. Standard fairness measures and their identifiability under selective sampling in screening applications. ✓ identified from labeled corpus; • identified with additional unlabeled data (known population group sizes); × not identified without population-level labelled data. TP, FP, TN, FN are the counts of the contingency table. $N = \text{TP} + \text{FP} + \text{TN} + \text{FN}$ is the total group size of the incoming applicants. The total group size can be known even without knowing the ground-truth labels. $f \in [0, 1]$ are the predictive model scores and $a \in \{0, 1\}$ is the group membership indicator.

Measure	Statistic	Identified?	Basis for use
<i>Measures on positively predicted cases</i>			
Statistical parity	$(\text{FP} + \text{TP})/N$	•	Disparate impact law; 80% rule [7, 38]; [35, 52]
Predictive Parity, Precision	$\text{TP}/(\text{FP} + \text{TP})$	✓	Information retrieval [59]; clinical [2]; [21]
False Discovery Rate	$\text{FP}/(\text{FP} + \text{TP})$	✓	Multiple testing [10]; information retrieval [59]; [21, 26]
<i>Measures on negatives</i>			
Predictive Equality, False Positive Rate	$\text{FP}/(\text{FP} + \text{TN})$	×	COMPAS audit [5, 21]; harm to innocents [25]
True Negative Rate, Specificity	$\text{TN}/(\text{FP} + \text{TN})$	×	Clinical diagnostics, screening [1]; [43]
False Omission Rate	$\text{FN}/(\text{FN} + \text{TN})$	×	[33]
Treatment Equality	FN/FP	×	[11]; not expressable as probability
<i>Measures on true outcomes</i>			
Equal Opportunity, True Positive Rate	$\text{TP}/(\text{TP} + \text{FN})$	×	Recall in ML [59]; benefit to qualified [35, 43]
False Negative Rate	$\text{FN}/(\text{TP} + \text{FN})$	×	Clinical screening [1]; wrongful denial [7, 43]
Equalised Odds, Separation	TPR & FPR	×	[6, 43]
Accuracy	$(\text{TP} + \text{TN})/N$	×	Standard in ML
<i>Other measures</i>			
Calibration / Sufficiency	$P(y = 1 f, a)$	×	Clinical prediction [75]; ML fairness [6, 21, 54]
FP Group	FP/N	•	City of Amsterdam custom [15]

Table 2. A toy example showing sample-based vs. population-level fairness measures under differential investigation for subgroups A and B and the scenario where training would have optimized for equality of FPR.

Measure	Training sample		Population		Match?
	A	B	A	B	
TP	60	34	60	34	yes
FP	140	136	140	136	yes
FN	5	8	40	66	no
TN	95	92	660	694	no
Precision	0.30	0.20	0.30	0.20	yes
FDR	0.70	0.80	0.70	0.80	yes
FP group	0.14	0.13	0.14	0.13	yes
FPR	0.60	0.60	0.18	0.16	no
FNR	0.08	0.19	0.40	0.66	no
TPR	0.92	0.81	0.60	0.34	no
FPR ratio A/B	1.00		1.07		no
FNR ratio A/B	0.41		0.61		no
TPR ratio A/B	1.14		1.76		no

Every measure marked as unidentified in Table 1 has either TN or FN in its denominator and TP or FP in its numerator. Since TP and FP are observed directly from the flagged pool, and $\widehat{FN}$ and $\widehat{TN}$ are recovered from Eq. (1), all such measures become identified.

The critical requirement is that the audit samples are uniformly at random from the uninvestigated pool. If the audit is itself conducted by the same selective process, as in the Amsterdam Prepilot, where caseworkers investigated additional cases according to their usual procedure, identifiability is not restored.

4 Failure mode II: reversal of disparity from training to deployment

Disparity reversal under case shift is the second failure mode that occurs in screening applications. Large differences in the prior probabilities of the classes between the training dataset and the population, combined with differential model accuracy across groups, lead to a situation in which a fairness intervention calibrated on the training corpus reverses the disparity at deployment. This failure mode occurs independently of the non-identifiability of the fairness measures discussed, but non-identifiability makes it harder to diagnose. For clarity of exposition, we analyze this using FDR, the only fully identified measure of harm to individuals, so that non-identifiability does not confound the argument.

A model can be more accurate for one demographic group than another, a phenomenon we refer to as differential accuracy. From a classification perspective, we can assume that accuracy is defined by the ranking of cases based on the model scores. A perfectly accurate classifier would always rank all fraud before all non-fraud cases. We call this an oracle. On the other hand, a random classifier would assign a rank to each case as a uniformly random

mixture of fraud and non-fraud cases. There would be classifiers between random and oracle. The closer a classifier is to the oracle, the more accurate it is.

Differential accuracy across demographic groups has been documented empirically in several high-stakes domains [17, 65]. Differential accuracy can arise from biased outcome labels, from proxy features that correlate with the outcome for some groups but not others, or from demographic skew in the training corpus. The last source is particularly relevant here. If one group makes a large share of the training cases, the model primarily learns the patterns of that group and can be expected, by standard learning-theoretic arguments, to generalize more accurately within it due to a larger training sample [73, 77]. In the case we study, Dutch applicants account for 85% of training cases, against an estimated 50–60% of the true population; in this case, differential accuracy is structurally expected.

Let us consider what happens to FDR under differential accuracy. Suppose that the underlying base rate of fraud is the same in both groups and the model flags the same proportion of applicants from both groups using the same model threshold (no fairness interventions). If the model is differentially accurate and a single threshold is applied to both groups, FDR will differ between them. The group for which the model is less accurate will have a higher FDR, because its flagged pool contains proportionally fewer true fraud cases.

A fairness intervention that equalizes FDR on the training corpus must therefore apply different effective thresholds to the two groups. Even if the nominal threshold is the same and group membership is excluded from the input features, proxy features correlated with group membership carry the signal indirectly and effectively implement different group thresholds when FDR parity is optimized for under differential accuracy.

Consider what happens under case shift, when the fraud prevalence changes drastically from training to deployment. Suppose that a threshold is fixed during the training, and the same model is applied to the population at deployment. Let us assume that the classifier accuracy (ability to detect fraud correctly) does not change from training to deployment. However, the deployment cases overall can be more difficult, less difficult, or the same difficulty as the training cases. Let us decompose.

At training, the pool contains a high proportion of fraudsters. Many people have high scores simply because many people are fraudsters, and fraudsters tend to score high. At deployment, the pool contains proportionally fewer fraudsters. Now, most high-scoring people are not fraudsters but innocent people who happen to have suspicious-looking features. Since the model thresholds were calibrated for a world with much more fraud, the model is too aggressive for the true population. While it still flags everyone above the fixed threshold, now the majority of flagged people are innocent. This raises FDR.

The more accurate the model, the stronger the effect. Since the model is not an oracle, the flagging threshold sits at a point where both false positives and false negatives occur. A more accurate model has less overlap, since the two distributions are more separated. When prevalence drops, the thin tail of the positive distribution above the threshold shrinks rapidly, while the thick negative tail above the threshold stays. The ratio collapses faster than for a less accurate model, where both distributions have more mass above the threshold. Thus, the FDR of a more accurate model is more sensitive to changes in prevalence.

Now, a data scientist trains an initial model without any intervention. Suppose the model is more accurate for group A than for group B, even though there is no true underlying disparity in the base rates. The tests on the training corpus would show a higher FDR for group B due to lower accuracy on that group. Suppose the data

scientist implements a fairness intervention (e.g., reweighing the training instances [53]) to balance FDR and succeeds in doing so.

While the result is a balance between FDR in the training data, the models are still differentially accurate. If we now keep the thresholds fixed and deploy to the true population, FDR will increase, but FDR for the more accurate model will increase more. This then leads to the reversal of disparity, and group A is now at a disadvantage despite the fairness intervention just performed.

We illustrate the mechanism with the following model. Let $z \sim \text{Uniform}[0, 1]$ denote the underlying ground-truth propensity of fraud. Suppose that real frauds are those cases where $z \geq \pi$, where π is the base rate of fraud in a population.

A model produces an evidence score as $f = \alpha z + (1 - \alpha) \varepsilon$, where $\varepsilon \sim \text{Uniform}[0, 1]$ and is independent of z . The higher the model score, the higher is the suspicion of fraud.

The weight $\alpha \in [0, 1]$ is the weight of signal over noise, and in this simulation, it acts as a proxy for model accuracy. If $\alpha = 1$, then the model perfectly recovers the propensity, that is, the oracle. If $\alpha = 0$, the model output is pure noise, and it implements a random classifier. Anything in between implements a classifier that is better than chance, but worse than the oracle. This means that there will be a region of model scores where the scores for true frauds and non-frauds overlap. A case is flagged for suspected fraud if $f > t$, where t is an externally fixed threshold.

We simulate two groups, A and B, with the same prevalence but different model accuracy, $\alpha_A = 0.8$ and $\alpha_B = 0.4$. The training corpus has prevalence $\pi^{\text{train}} = 0.5$. The deployment population has prevalence $\pi = 0.1$.

A simulated fairness intervention is illustrated in Figure 2. The thresholds for both groups were selected so that each group achieved the same FDR (0.1). The models are calibrated on the training data so that there is no apparent disparity between groups (Figure 2(a)). When the model is deployed on the population using the same thresholds, FDR rises for both groups, but it rises more for group A (Figure 2(b)). The group for which the model was initially more accurate fails more often. Disparity between the groups persists in the population data despite the fairness intervention applied to the training data. The reversal is a structural consequence of calibrating on an enriched corpus and deploying at the true population prevalence. This illustrates a case shift in action and mimics the reversal of disparity that happened in the Amsterdam case.

While there is no easy solution to this problem, rebalancing the training data to reflect the expected population prevalence, similar to reweighing for fairness interventions, could help. As our focus here is analytical, on formalizing two highly consequential modes of failure in fairness interventions, we leave a design of detailed algorithmic solutions open for future work.

5 The City of Amsterdam welfare fraud detection case

Background. Between 2021 and 2023, the City of Amsterdam developed and briefly deployed a system to score welfare applications for investigation regarding potential fraud [4, 15, 42, 78]. After the Pre-pilot evaluation found the original model flagged non-Western applicants disproportionately, the city applied a fairness intervention, retrained the model, declared the bias corrected, and proceeded to a live Pilot. The Pilot showed the disparity had reversed. We reconstruct this from the confusion matrices and code released for Lighthouse Reports [15].

The city used a state-of-the-art model called Explainable Boosting Machine [58, 64]. Their training corpus consisted of ~ 3400 applications that were flagged manually by caseworkers and investigated for potential fraud

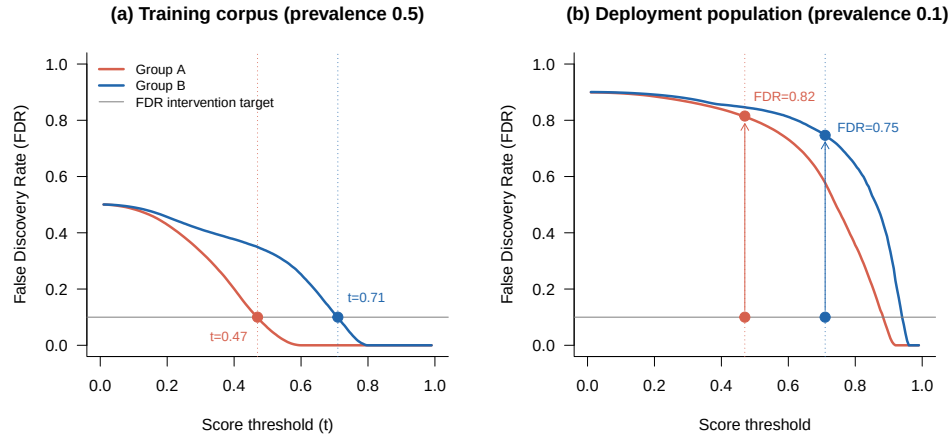


Fig. 2. Simulation to illustrate reversal of disparity: (a) training corpus, (b) deployment on population.

following the regular procedures of the city. Based on the investigation outcome, the cases were assigned ground-truth labels (fraud or non-fraud). For training the model, the city used fourteen behavioral and financial features. Potentially sensitive features, such as nationality, gender, or age, were not included in the model training.

The project proceeded in three stages. In the Training stage, an initial model was fit optimizing for accuracy (AUC) with no fairness constraints. During the development, the city consulted its Welfare Advisory Council, a body of citizens and interest group representatives. The Council formally recommended stopping the project, citing discriminatory risk from sensitive variables, disproportionate use of personal data, and lack of democratic oversight [78]. The development team removed potentially discriminatory variables and proceeded to the Prepilot.

In the Prepilot stage, cases continued to be flagged manually, but model scores were recorded in parallel for testing purposes. An additional sample flagged by the model but not by caseworkers was also investigated manually and labeled. Of all flagged cases, approximately 39% overlapped between the model and caseworkers, as we found from their code [15]. The city considered false positive rate parity as a desired fairness measure but ruled it out as true and false negatives were structurally absent [3]. The city thus devised a tailored fairness measure. The Prepilot evaluation showed that the model disproportionately flagged non-Western applicants. To correct for this, the city applied AIF360 reweighing [9, 53] to the original training corpus and retrained the model until the disparity was balanced as measured on the Prepilot labeled dataset. The city's external technical advisor warned in writing that reweighing might reduce bias against one demographic group while creating new disparities for others [78].

In the Pilot stage, which ran for three months, the model primarily determined which applications were sent for investigation. Caseworkers conducted the actual investigations. It turned out that the outcome codes used by enforcement officers to classify cases were ambiguous and interpretable in multiple ways, undermining label quality, and several demographic groups contained too few cases for statistically reliable conclusions. The city therefore could not fully determine whether the Pilot model was discriminatory. Where estimates were possible, however, the disparity had reversed. Applicants with Dutch nationality were substantially more likely to be wrongly flagged than non-Dutch applicants [3]. In addition, women now were more likely to be wrongly flagged than men.

In November 2023, following a formal recommendation by the external consultant calling for termination on grounds of inevitable discrimination risk and insufficient demonstrated effectiveness [78], Deputy Mayor for Social Affairs halted further development, citing the inability to guarantee prevention of unintentional discrimination.

Fairness measure. The city conducted their own bias analysis [3], which outlines a thorough consideration of fairness measure choice. Using the Aequitas fairness tree [71], the city considered three candidate measures. False discovery rate parity was rejected as too focused on model efficiency rather than on fairness to applicants. False positive rate parity was explicitly ruled out since only positively flagged cases were investigated, true and false negatives were structurally absent, making it impossible to estimate FPR from the available data. This maps to our non-identifiability argument. The city noted this, stating that applications that were not investigated cannot be assumed to be non-fraudulent. Thus, to measure fairness meaningfully, the city defined own custom measure. The city defined FP Group Size, $FPG = FP^* / N$, where FP^* was an extrapolated false-positive count and N was the group size. The extrapolation scaled observed false positives from labeled model-only flags to total model-only flags, assuming the unlabeled cases had the same outcome distribution as the labeled ones. The ratio between FPG for non-Westerns vs. Westerns was reported as 1.725 for the initial model and 0.997 for the reweighed model.

FPG relates to FPR and FDR as follows. For FPR, let $n^{neg} = (1 - \pi)N$ denote the count of negatives in the group, where π is the fraud prevalence. Then $FPR = FP / n^{neg}$. Substituting gives $FPG = FP / N = \frac{FP}{n^{neg} / (1 - \pi)} = FPR(1 - \pi)$. For FDR, let $r = (FP + TP) / N$ denote the flagging rate for the group as measured in the Statistical Parity. Substituting gives $FPG = FP / N = \frac{FP}{FP + TP} \frac{FP + TP}{N} = FDRr$. FPG therefore equals FPR scaled by $(1 - \pi)$ and FDR scaled by r . FPG is equal to FPR and FDR only if prevalence and flagging rates are equal across groups correspondingly.

Turning to identifiability, each measure has a different status in this case. FPG is identified but requires knowledge of the total group sizes N . Since we do not have access to N in the released data, we cannot reconstruct it. FPR requires true and false negatives in its denominator and is unidentified under selective labeling at every stage. FDR is identified only when the flagged cases are determined by the model itself, so that true positives and true negatives reflect the model's decisions. This condition holds in the Amsterdam Training stage, where the model was fit, and in the Pilot stage, where the model determined which cases to flag for investigation. FDR, however, was not identified in the Pre-pilot stage, where cases were flagged primarily by caseworkers, and some model-flagged cases were left unlabeled. In the following analysis, we thus treat FDR as meaningful only for the Training and Pilot stages.

The reversal of disparity. In the following reconstruction we use the Dutch/non-Dutch split throughout because the Western/non-Western split is partially suppressed in the released data due to small group sizes. Around 95% of Western nationals in the corpus are Dutch, so the splits are closely related.

Table 3 shows FDR and FPR by stage and model. As established above, we treat FDR as the identified measure and base the analysis on it. FPR is included in the table for reference but is unidentified and not analyzed.

The key comparison is between the training and pilot stages, where FDR is identified. The initial model (BR), without any fairness intervention, shows an FDR ratio (F/D) of 1.08 at Training, indicating a mild disadvantage for non-Dutch applicants. At the Pilot, where prevalence drops drastically, the ratio reverses to 0.84, now indicating a disadvantage for Dutch applicants. The absolute FDR increases for both groups, but more so for Dutch applicants, consistent with the mechanism we derived in the analytical sections. This reversal occurs due to a case shift, even without any fairness intervention.

The reweighed model (AR) tells the same story. At Training, the intervention achieves near-parity with FDR ratio 1.04. At the Pilot, the ratio reverses to 0.79, again disadvantaging Dutch applicants.

Table 3. FDR and FPR by stage and model, Dutch (D)/non-Dutch (F) split. BR – the initial model; AR – reweighed model. π is the prevalence of fraud in the dataset. FDR – false discovery rate, FPR – false positive rate. Bold indicates the identified rates.

Stage	Model	N	π	FDR			FPR		
				D	F	ratio F/D	D	F	ratio F/D
Train	BR	2794	0.55	0.33	0.36	1.08	0.24	0.25	1.03
Train	AR	2794	0.52	0.31	0.33	1.04	0.19	0.21	1.10
Prepilot	BR	1221	0.47	0.40	0.33	0.84	0.26	0.44	1.70
Prepilot	AR	1221	0.60	0.41	0.23	0.57	0.22	0.22	0.97
Pilot	BR	271	0.46	0.53	0.45	0.84	0.66	0.63	0.96
Pilot	AR	271	0.54	0.57	0.44	0.79	0.73	0.59	0.81

These results validate our analytical insights and show that disparity reversal is not an artifact of a flawed intervention. It is a structurally prescribed consequence of calibrating on an enriched corpus and deploying on the true population, embedded in the learning task setup.

6 A taxonomy of screening applications

While we analyzed the failure modes through the example of welfare fraud detection, where a positive flag is adverse for the individual and ground-truth labels are available only for flagged cases, the same structural challenges arise across a broader class of screening applications. Screening settings can be distinguished along two dimensions: whether a positive prediction is adverse or beneficial for the individual, and whether ground-truth labels are observed for positively or negatively screened cases. By convention, the positive label denotes the rare class, such as fraud or a hiring decision, while the negative label denotes the common class, such as no fraud or no hiring. The four types that result from crossing these two dimensions are summarized in Table 4.

Types B+ and G+ share the same identifiability structure: labels exist only for screened-in cases, so true and false negatives are absent, and the corpus has a higher prevalence than the true population. Both failure modes, non-identifiability and disparity reversal, therefore apply directly to both types. What distinguishes them is normative: in B+, false positives harm more (wrongful investigation), while in G+, false negatives harm more (wrongful denial). In neither case, FPR nor FNR, are identified. FDR and precision are the identified measures of harm and can be meaningfully used to guide fairness interventions in both types.

Types B- and G- have the opposite labeling structure: ground-truth labels are observed only for screened-out cases, since the adverse outcome can only materialize following a negative decision (e.g., reoffending requires release, default requires credit being granted). True and false negatives are therefore observed, while true and false positives are absent. Non-identifiability applies to a different set of measures than in B+ and G+, and prevalence is suppressed in the corpus rather than enriched, with fewer positives in the training data than in the true population. Both failure modes apply, but with reversed identifiability structure and reversed direction of prevalence distortion.

G- is practically rare. The only case we identified is organ transplantation, where the prediction target is urgency. Being ranked high is good for the patient, yet ground-truth urgency is observed only for those who did not receive

Table 4. Screening applications of algorithmic fairness by type.

Type	Positive prediction	Labels from	Prevalence	Applications
B+	bad	flagged +	amplified ↑	Welfare fraud detection [15, 37, 42, 76]
				Childcare benefit fraud detection [31, 76]
				Tax audit selection [14, 36]
				Child protective services [22, 37]
B-	bad	released -	suppressed ↓	Criminal risk assessment [5, 34]
				Credit risk assessment [8]
G+	good	approved +	amplified ↑	CV screening for hiring [30]
				University admissions [12, 81, 86]
				Clinical trial eligibility screening [19]
G-	good	rejected -	suppressed ↓	Ranking patients for organ transplantation [29, 60]

an organ and remained on the waiting list. B- is more common, with criminal recidivism risk assessment being the canonical example: reoffending is observed only for defendants who were released.

The infamous COMPAS case is type B-. High recidivism risk scores lead to detention, and detained defendants cannot reoffend, so outcome labels exist only for released defendants. The ProPublica/Northpointe dispute [5] therefore cannot be resolved on the available data. ProPublica’s FPR claim requires knowing the reoffending counterfactual for detained defendants, which is structurally unobservable, making FPR an unidentified measure in this setting. Northpointe’s calibration claim, by contrast, was computed within the released population and is identified. Chouldechova et al. [21] showed that both claims can simultaneously be true when base rates differ across groups, but the more fundamental problem is that the two sides were computing measures with different identifiability properties from the same dataset, without either making this explicit.

The COMPAS case is one of many high-stakes screening applications. The typology shows that all four types face conceptually identical structural challenges. While the specific set of identified and unidentified measures varies across types, the failure modes are embedded in the task structure itself, not in any particular implementation choice. Settings where labels are representative of the full population, as in image classification or natural language processing, are comparatively rare among the most consequential and fairness-sensitive applications of algorithmic systems. Selective labeling should therefore be treated as the default assumption in fairness-aware machine learning rather than a special case, and fairness measures, interventions, and their statistical guarantees should be designed to reflect this structure from the outset.

7 Related work

Several lines of work relate to different aspects of our study.

Selective labeling and sample bias. The selective labeling problem in algorithmic decision making was formalized by Lakkaraju et al. [55]. They showed that evaluating learned rules against historical decisions conflates the algorithmic performance with that of the human whose decisions generated the labels. Coston et al. [27] developed a counterfactual risk assessment framework for pretrial detention under partial observability, and De

et al. [32] studied the feedback loop between lending decisions and observed repayment outcomes. These papers address prediction under selective labeling but do not analyze its consequences for fairness metric estimation.

The statistical foundations of our setting are well established. The problem structure is related to sample selection bias in econometrics [46] and to censoring in survival analysis [28], but differs from both. In our setting, selection is endogenous to the decision process being modeled, and the missing outcomes are not recoverable by longer observation. Our setting is also distinct from covariate shift [68] and the class imbalance corrections proposed for it [84]. In their setting, the selection mechanism depends on the input features, whereas in our screening applications setting, it depends on the true outcome. Standard importance weighting treatments, therefore, do not directly apply to our setting.

More generally, the case shift that we defined here in the context of algorithmic fairness interventions is a specific instance of missing not at random [56, 69], with outcome-dependent sampling [16, 67]. Our contribution is to connect this structure to Fairness metric estimation explicitly and to analyze the conditions under which it produces measure non-identifiability and disparity reversal.

Fairness measure incompatibility and non-identifiability. A substantial literature highlights that popular fairness measures are mutually incompatible. Chouldechova et al. [21] proved that FPR parity, FNR parity, and predictive parity are simultaneously achievable only when prevalence is equal across groups. Kleinberg et al. [54] derived analogous impossibility results for calibration and balance. Corbett-Davies et al. [25] showed that enforcing classification parity typically produces Pareto-dominated decision policies. Defrance and De Bie [33] characterized the maximal sets of jointly satisfiable fairness criteria. Zhao and Gordon [85] showed information-theoretic tradeoffs between many fairness measures and accuracy. Loi et al. [57] provided a normative analysis of calibration versus separation, concluding that which criterion fairness demands depends on the decision structure.

Our contribution adds a prior question to these incompatibility results. In the most consequential real-world settings, those where the incompatibility debate has been most heated, many of the measures involved cannot be estimated from the available data. Our results show that identifiability must be resolved before compatibility is even considered, since if the measure is non-identifiable, compatibility does not make sense anyway.

Non-identifiability of fairness measures has been considered when the protected attribute is unobserved [20, 50], with partial identification bounds proposed as a remedy. Our setting here is orthogonal. We consider the situation in which the protected class is observed, but outcome labels are missing for negatively screened cases. To our knowledge, the systematic identifiability analysis under selective labelling in fairness contexts has not been addressed before.

Differential accuracy across demographic groups. Differential model accuracy across demographic groups is well documented empirically. Buolamwini et al. [17] showed that commercial face recognition systems have substantially higher error rates for darker-skinned women. Obermeyer et al. [65] showed that a widely used health risk score was systematically less accurate for black patients due to a biased training label. Coston et al. [27] and Mishler et al. [62] analyzed differential predictive performance under selective labeling in criminal justice settings.

The psychometric literature has a longer tradition of studying the same phenomenon under the names of differential item functioning [47] and differential prediction [24]. More recently, Hashimoto et al. [44] and Sagawa et al. [70] showed that standard empirical risk minimization tends to produce models that perform well on average but fail disproportionately on minority subgroups, with the gap widening over time.

Here, we consider differential accuracy from a particular perspective. We derive the conditions under which differential accuracy interacts with case shift to produce a reversal of fairness measure disparities between training and deployment, rather than merely a widening of the accuracy gap.

Fairness and distributional shift. The mismatch between training and deployment distributions has been widely studied under the framework of dataset shift [68], concept drift [40], and transfer learning [66]. Singh et al. [74] studied fairness interventions under covariate shift. Kallus et al. [51] showed that fairness interventions can redistribute rather than reduce harm when residual confounding is present.

Our identified case shift differs from all of these in a fundamental respect. In our setting, the underlying population distribution remains unchanged. The mismatch between training and deployment is endogenously induced by the screening process itself and is therefore analytically predictable from the structure of the task. This distinguishes our setting from covariate shift, label shift, and confounding-induced redistribution, in which the population or data-generating process is assumed to have changed or to be incorrectly specified and must be detected and corrected post hoc. It is therefore an additional orthogonal challenge, and we argue that case shift takes priority because it is not merely a possibility but already embedded in the task setting, and its consequences, if unaccounted for, can be much more severe.

8 Implications for responsible AI practice

The Amsterdam case that motivated our study was not a story of negligence in fairness. The city consulted external experts, published the code, excluded demographic features from the model, applied a classical bias-correction procedure from a widely used library, and ran a dedicated prepilot evaluation. Their own analysts even flagged in their notebook the specific risk that unlabelled model-flagged cases were compositionally different from labeled ones. As evident from the report [3], the city also found out that the measure they would like to target was not possible to measure meaningfully. They chose a measure that is possible to measure. However, their fairness intervention still made things worse.

The failure was structural, as the evaluation framework the city used could not distinguish a genuine fairness improvement from an artifact of case shift. Our analysis highlights that despite the proliferation of responsible AI toolkits [9, 13, 49, 71, 82], none have yet included the methodologies needed to evaluate fairness faithfully in selective labeling settings, which we argue are among the most, if not the most, prevalent settings in fairness applications. Our analysis thus motivates three concrete recommendations for practice.

First, fairness claims should not be based on measures where denominators depend on unobserved outcomes. The most widely used measures, such as FPR, FNR, Equalized Odds, and Equal Opportunity, even motivated normatively [35, 39, 43], are not identified statistically under selective labeling. Calibrating a fairness intervention to an unidentified measure provides no guarantee of parity at the population level and, as we have shown, it can even reverse the disparity it was designed to correct. FDR and precision are identified in screening applications and should be used primarily.

Second, any fairness intervention should be anchored in training data in which prevalence and demographic composition match those of the expected deployment population. The simplest approach is to apply the same reweighting protocol used for fairness [53] to first correct for prevalence mismatch and demographic skew before any fairness intervention is applied.

Third, differential model accuracy across demographic groups should be treated explicitly rather than as an afterthought. This is the most challenging recommendation because accuracy itself is only partially measurable under selective labeling, where precision is identified, but recall is not identified. However, even beyond measurement, there are cases where differential accuracy cannot be eliminated, such as when the minority group is too small to train on with equal fidelity, regardless of intervention design. Where that is the case, the differential accuracy should be acknowledged openly and factored into the choice of fairness criterion. Most critically, any fairness intervention should then be calibrated on data whose class balance reflects the expected deployment population.

Beyond the technical recommendations, several limitations of our analysis need to be acknowledged.

In our setup, we assumed a binary classification task, but in practice, the mapping from raw investigation results to a binary fraud label is not unique. The Amsterdam system encountered many outcome types, including different categories of fraud, procedural findings, and inconclusive results. This mapping to the binary training labels discards broader structures. The binarisation choice is open to examining.

We also assumed that labels accurately reflect ground truth. In welfare fraud systems, labels are the product of caseworker judgment. If caseworkers are more likely to confirm fraud for one group than another under the same evidence explanations [48, 65], the training labels are systematically biased independently of case shift, and the two problems compound each other.

We further assumed that the underlying population distribution is stable over time and that the shift from the training set to the population is solely due to selective sampling. Genuine concept drift [40] is orthogonal to case shift and may amplify it, and our conditions for reversal should be interpreted accordingly.

Finally, our setup, which permits different investigation thresholds for groups while being statistically motivated, raises legal questions in many jurisdictions [80, 83]. Applying different flagging rates to different demographic groups may itself be prohibited discrimination, even when the intent is to equalize fairness.

9 Conclusion

We pointed out that, despite the maturity of fairness-aware machine learning, several fundamental algorithmic challenges remain unresolved, particularly in applications where fairness matters most.

Screening systems used to flag cases for investigation or approval generate outcome labels only for cases that a prior process has already selected. This induces a mandatory distribution shift from training to deployment, which we term a case shift. Under the case shift, the most widely used fairness measures become unidentified. This implies that they cannot be reliably estimated from available data and therefore cannot faithfully guide fairness interventions. And even measures that remain identified can amplify disparity or reverse its direction at deployment.

The City of Amsterdam welfare fraud detection system provides empirical confirmation of these failures. Our analysis shows that the breakdown of this system was prescribed by the learning task setup.

Our findings do not imply that fair automation in screening applications is impossible. We argue that achieving fairness from an algorithmic perspective requires further progress on two fronts: a rigorous characterization of the behavioral conditions under which existing fairness measures and interventions remain valid under selective labeling, and the development and tailoring of algorithmic methodologies that operate natively under case shift. As selective labeling is not an anomaly or an implementation flaw but a structural feature of any screening automation, and fairness-aware machine learning methodologies must be built to that reality.

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